北京郵電大學

Queen Mary University of London

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# EBU5376 B

**Joint Programme Examinations 2022/23**

**EBU5376 Digital Signal Processing**

**Paper B**

**Time allowed 2 hours**

**Answer ALL questions**

**Complete the information below about yourself very carefully.**

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| **QM student number** | | | | | | | | | | |
| **BUPT student number** | | | | | | | | | | |
| **Class number** | | | | | | | | | | |

**NOT allowed: electronic calculators and electronic dictionaries.**

**INSTRUCTIONS**

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

**Examiners**

Dr Yixuan Zou, Andy Watson

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Filename: 2223_EBU5376_B No answer book required

# Instructions

**Before the start of the examination**

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

**During the examination**

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

**At the end of the examination**

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave

**Question 1**

a) This question is about the Laplace transform. **[12 marks]**

i) State the formula of Laplace transform on a continuous function $x(t)$. (2 marks)

ii) Perform Laplace transform to the following function (10 marks)

$$x(t) = 2\cos(3t)\,u(t).$$

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| | **12 marks** |

b) This question is about the z-transform, which is another transformation between real and complex domain. **[13 marks]**

i) State the relationship between the z-plane and the s-plane. (4 marks)

ii) Perform inverse z-transform to the following sequence (9 marks)

$$X(z) = \frac{-2z^2 - \frac{2}{3}z}{z^2 + \frac{2}{3}z - \frac{1}{3}}$$

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**Question 2**

This question is about the family of Fourier Transform, which consists of Fourier Series (FS), Fourier Transform (FT), Discrete Time Fourier Transform (DTFT), and Discrete Fourier Transform (DFT).

a) State the assumptions on the input signals by the four members of the Fourier Transform family. **[8 marks]**

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| | **8 marks** |

b) State and prove the time-shift property of Fourier Transform. **[8 marks]**

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| | **9 marks** |

**Question marking**: $\frac{\quad}{8} + \frac{\quad}{8} + \frac{\quad}{9} = \frac{\quad}{25}$

**Question 3**

This question is about the Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT).

a) DFT is the only member of the Fourier Transform family that can be implemented by computers. **[8 marks]**

i) State the order of complexity of an N-point DFT. (2 marks)

ii) Perform DFT to the following sequence:

$$x[n] = [1, -j, -1 - j, 2j]$$

(6 marks)

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| | 8 marks |

b) FFT is the low-complexity algorithm for computing DFT. **[17 marks]**

i) State the order of complexity of an N-point FFT. (2 marks)

ii) Using a butterfly diagram, apply decimation in time (DIT)-FFT to the following sequence

$$x[n] = [1, -j, -1 - j, 2j]$$

(6 marks)

~~iii) FFT reduces the computational complexity of DFT by utilizing the divide and conquer approach. Show that an N-point DFT can be decomposed into the summation of two $\frac{N}{2}$ point DFTs. (9 marks)~~

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| | **17 marks** |

**Question marking**: $\frac{\quad}{8} + \frac{\quad}{17} = \frac{\quad}{25}$

**Question 4**

a) There are two types of digital filters, namely, Infinite Impulse Response (IIR) and Finite Impulse Response (FIR). The Generic Digital Filter Transfer Function is:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{i=0}^{M} b_i z^{-i}}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

State and explain the condition for a filter to be FIR and comment on the stability of FIR filters. **[4 marks]**

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| | 4 marks |

b) The difference equation of a particular non-recursive filter is: **[8 marks]**

$$y[n] = 0.6 + 0.5x[n]$$

i) Draw its signal flow graph. (2 marks)

ii) Determine the filter's transfer function. (1 mark)

iii) Hence calculate the frequency response amplitudes at frequencies of 0, π/4 and π/2 rad/sample interval. Full working must be shown. (5 marks)

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| | **8 marks** |

c) The following transfer function is that of a recursive filter:

$$H(z) = \frac{(z+1)(z-1)}{(z-0.9)(z+0.6)}$$

**[13 marks]**

i) Draw its poles and zeros on the z-plane and comment on its relative stability.

(8 marks)

ii) Determine the filter's difference equation. (5 marks)

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